

SETS. VENN DIAGRAMS. FUNCTIONS – MTH 101

Session 1

MTH 101

LEARNING OBJECTIVES

Upon completion of this unit, students should be able to:

- Describe operations on sets using Venn diagrams.
- Perform operations on one or two sets such as unions, intersections, and complements.
- Use the formula of the cardinal of the union of two finite sets to solve basic counting problems.
- Use Venn diagrams to prove equalities between sets.
- Apply the formula of the cardinal of the union of two finite sets to real-world situations.
- Recognize functions between two sets.

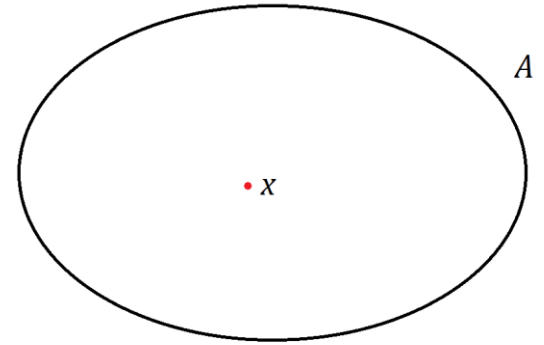
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WHAT IS A SET?

A **set** is a well-defined collection of distinct objects which are called its members or elements.

If x is a member of a set A , we write $x \in A$. If x is not a member of a set A , we write $x \notin A$.

Sets can be described using the roster or enumeration method, the set-builder notation, or by semantic definition.



$A = \{\dots, -2, -1, 0, 1, 2, \dots\}$ (Roster notation)

$A = \{\pm x \mid x \text{ is a whole number}\}$ (Set-builder notation)

A is the set of whole numbers and their negatives (Semantic definition)

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FINITE SETS AND CARDINALITY

A **finite set** is one for which it is possible to list all its elements. On the other hand, an **infinite set** is a set with a never-ending number of elements.

The cardinal value or **cardinality** of a finite set is the number of elements in the set. For a finite set E , it is denoted as $n(E)$ or $|E|$.



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EXAMPLES

The set of all lowercase letters of the English alphabet: $A = \{a, b, c, \dots, z\}$.

It is **finite** and has 26 elements so $|A| = 26$.

The set of all whole numbers: $B = \{0, 1, 2, 3, \dots\}$.

It is infinite.

The set of whole numbers between 50 and 100:

$C = \{x \mid x \text{ is a whole number and } 50 \leq x \leq 100\}$.

It is a finite set and $n(C) = 51$.



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EQUIVALENT AND EQUAL SETS

Two sets A and B are said to be **equivalent** if they have the same number of elements, that is, if they have the same cardinality). We write $A \sim B$.

Two sets A and B are said to be **equal** if they have precisely the same elements. We write $A = B$.



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EXAMPLES

- $\{h, a, p, y\} = \{a, p, h, y\}$
- $\{1, 2, 3, 4, 5\} \sim \{x, y, z, t, u\}$
- $\{1, 2, 3, \dots, 7\} \sim \{n^2 \mid n = 1, 2, \dots, 7\}$
- **Note:** Equal sets are equivalent but equivalent sets are not necessarily equal.

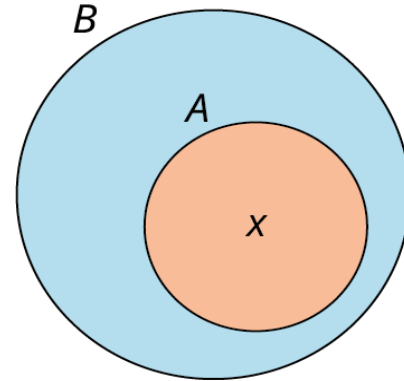
WHAT IS A SUBSET OF A SET?

Set A is a **subset** of set B if every member of set A is also a member of set B . We write $A \subseteq B$.

If $A \subseteq B$ and $A \neq B$, set A is said to be a **proper subset** of set B . We then write $A \subset B$.

Note:

1. Every set is also a subset of itself: $B \subseteq B$.
2. The empty set is a subset of every set: $\emptyset \subseteq B$.



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THE POWER SET

The **power set** of a set A is the set of all possible subsets of set A. It is denoted $P(A)$.

In the case of a finite set A, the number of subsets of A is equal to $2^{n(A)}$. Therefore, the cardinality of $P(A)$ is given by the formula:

$$n(P(A)) = 2^{n(A)}$$

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EXAMPLES

Let $B = \{1, 2, 3, 4, \dots, 10\}$.

$\{5\}$, $\{1, 3, 5\}$ and $\{7, 8\}$ are proper subsets of B .

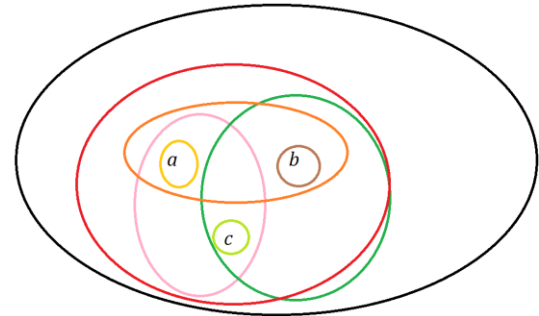
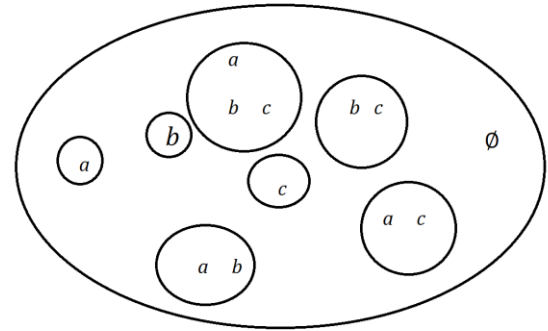
$\{1, -1, 2\}$ is not.

Let $A = \{a, b, c\}$. The possible subsets of A are:

$\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$, and $\{a, b, c\}$.

$$P(A) = \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

$$n(P(A)) = 8 = 2^3 = 2^{n(A)}$$



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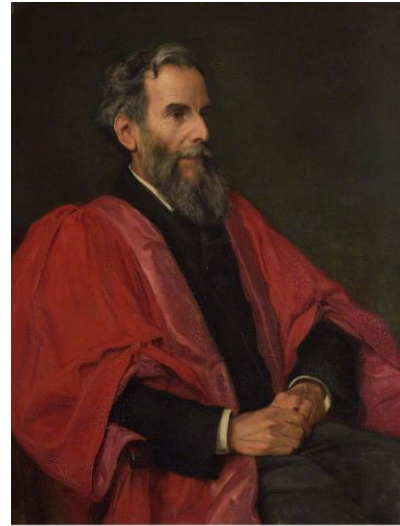
VENN DIAGRAMS

Venn diagrams are the graphical tools used to visualize and understand relationships between sets.

Venn diagrams are named after the mathematician John Venn (1843 – 1923).

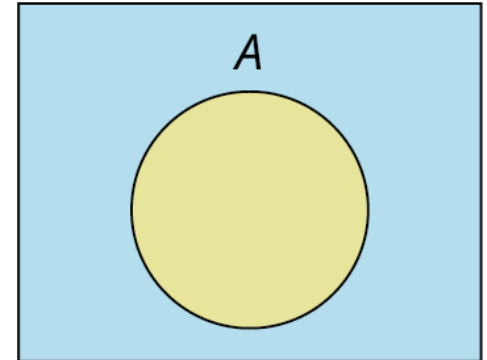
The entire set of elements under consideration is called the **universal set**. It is drawn as a rectangle. It is not unique as it varies from one problem to another.

Subsets within the universal set are drawn as circles, ovals, or other shapes, within the rectangle.



John Venn (1843 - 1923)

U

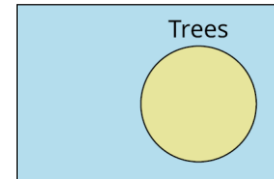


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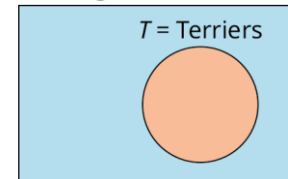
EXAMPLES

1. “All trees are plants”
2. “All terriers are dogs”
3. “All cats are lions”
4. “Triangles and squares are 2D figures”

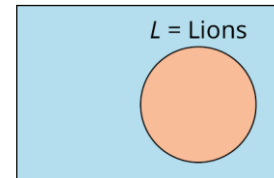
$U = \text{Plants}$



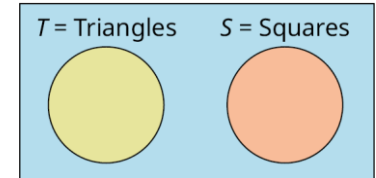
$U = \text{Dogs}$



$U = \text{Cats}$



$U = \text{2D Figures}$



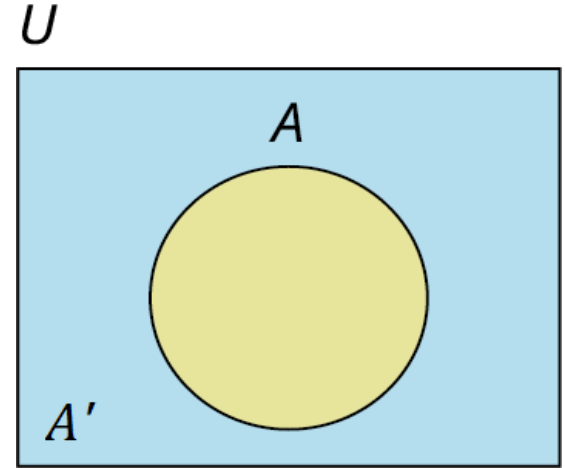
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COMPLEMENT OF A SET

The complement of a set A , denoted A' or A^c is the set consisting of all the elements in the universal set that are not in set A .

$$A^c = \{x \in U \mid x \notin A\}$$

The universal set U includes all the elements in set A , all the elements in set A' , and nothing else.



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EXAMPLE

Consider the universal set of digit numbers:

$$U = \{0, 1, 2, 3, \dots, 9\}$$

Consider the subsets $A = \{0, 1, 2\}$ and $B = \{5, 7, 9\}$ of U .

$$A' = \{3, 4, 5, 6, 7, 8, 9\}$$

$$B' = \{0, 1, 2, 3, 4, 6, 8\}$$

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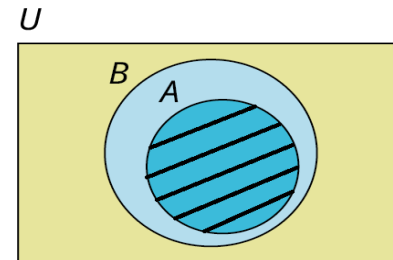
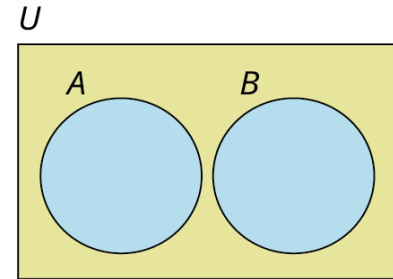
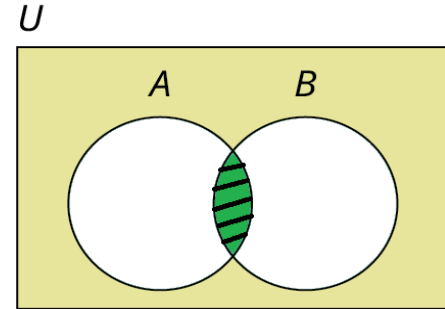
INTERSECTION OF SETS

The **intersection** of sets A and B, denoted as $A \cap B$ is the set of elements that are common to both sets A and B.

$$A \cap B = \{x \mid x \in A \text{ *and* } x \in B\}$$

Two sets are said to be **disjoint** if they share no element in common. We write $A \cap B = \emptyset$ if A and B are disjoint.

If $A \subseteq B$, then $A \cap B = A$.



EXAMPLES

- If $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 6, 9\}$ then

$$A \cap B = \{2, 4\}$$

- If $A = \{a, b, c, \dots\}$ and $B = \{f, a, i, l\}$ then

$$A \cap B = \{f, a, i, l\} = B$$

- If $A = \{1, 3, 5, 7, \dots\}$ and $B = \{2, 4, 6, 8, \dots\}$ then

$$A \cap B = \emptyset$$

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UNION OF SETS

The **union** of sets A and B, denoted as $A \cup B$, is the set of elements that belong to either set A or set B or both sets.

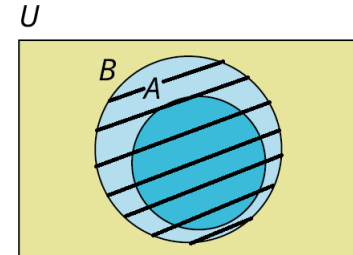
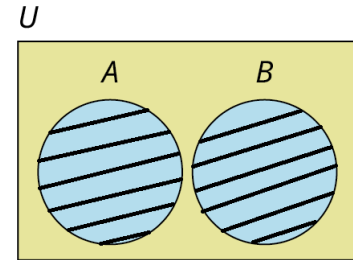
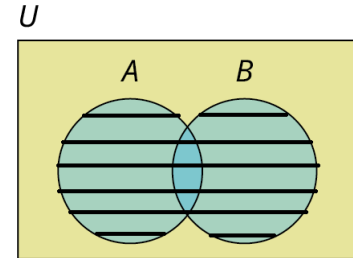
$$A \cup B = \{x \mid x \in A \text{ *or* } x \in B\}$$

For example:

If $A = \{1, 3, 5, 7, \dots\}$ and $B = \{2, 4, 6, \dots\}$ then $A \cup B = \{1, 2, 3, \dots\}$

If $A = \{c, u, p\}$ and $B = \{c, u, r, y\}$ then $A \cup B = \{c, u, p, r, y\}$.

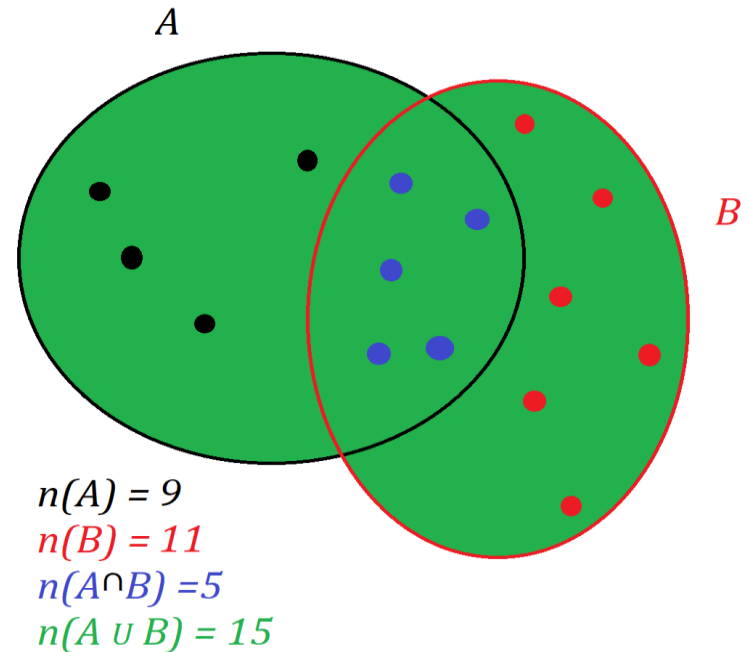
If $A \subseteq B$, then $A \cup B = B$.



CARDINALITY OF THE UNION OF TWO SETS

The **cardinality** of the union of two sets is the total number of elements in the set. It is found by adding the number of elements of set A to the number of elements of set B, then subtracting the elements in the intersection of set A and set B:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$



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OPERATIONS ON SETS

Let A and B be sets in a universal set U.

The complement of a set A is the set:

$$A^c = \{x \in U \mid x \notin A\}$$

The intersection of sets A and B is the set:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

The union of sets A and B is the set:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

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SOME MORE OPERATIONS

Let A and B be sets in the universal set U .

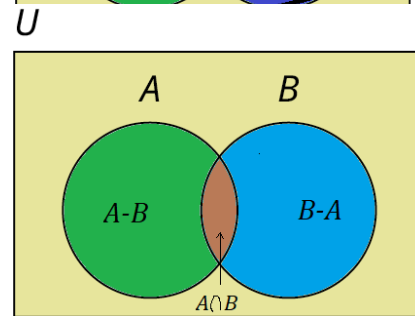
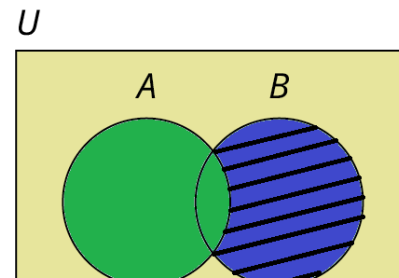
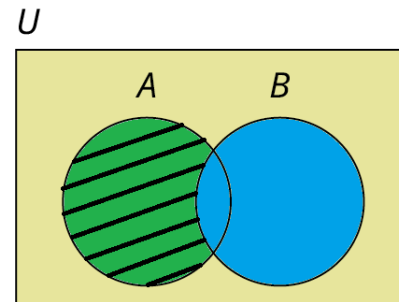
The **difference** of the sets A and B is the set of elements of A which belong to A but not B .

$$A - B = \{x \in A \mid x \notin B\}$$

The difference of the sets B and A is therefore:

$$B - A = \{x \in B \mid x \notin A\}$$

Note: $A \cup B = (A - B) \cup (B - A) \cup (A \cap B)$



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EXAMPLES

- If $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$ then:

$$A - B = \{1, 2\}$$

$$B - A = \{6, 7\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$$

$$A \cap B = \{3, 4, 5\}$$

- If set $A = \{0, 1, 2, 3, \dots\}$ and $B = \{0, 2, 4, 6, \dots\}$ then:

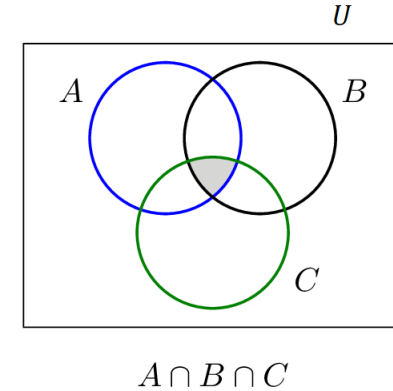
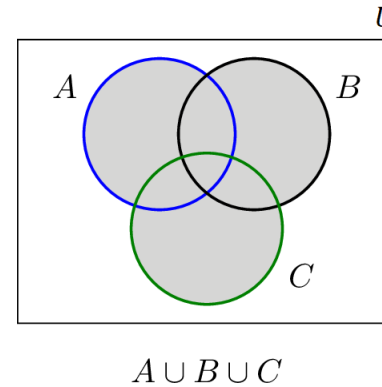
$$A - B = \{1, 3, 5, \dots\}$$

$$B - A = \emptyset$$



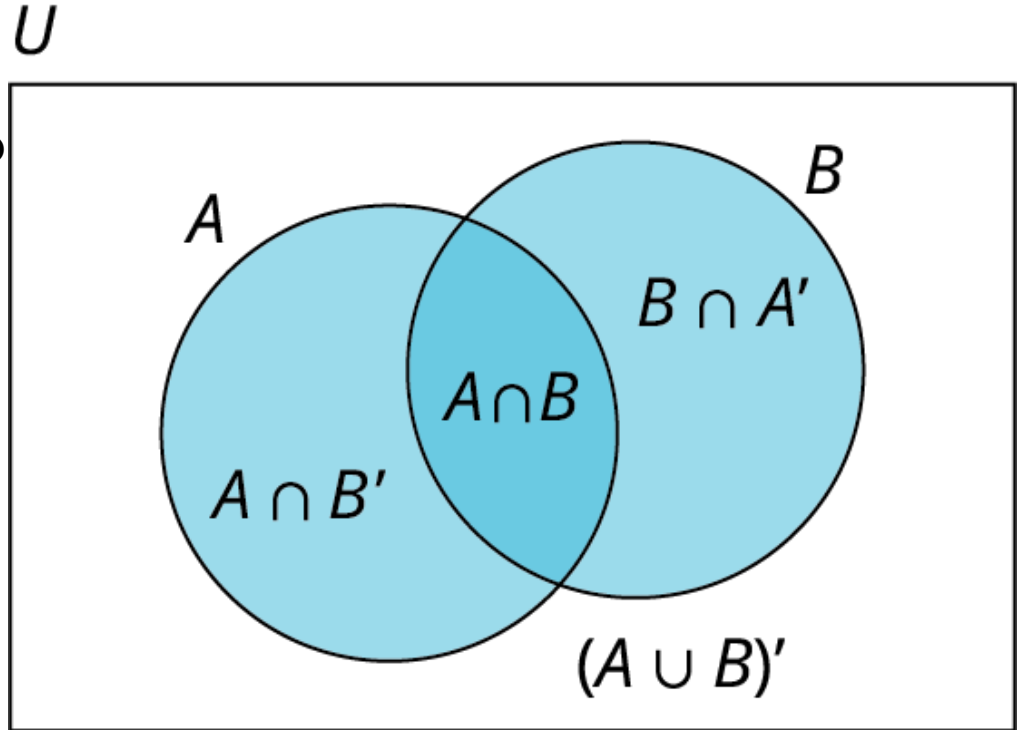
SOME PROPERTIES OF OPERATIONS ON SETS

- **Idempotency:** $A \cap A = A \cup A = A$
- **Commutativity:** $A \cup B = B \cup A$
 $A \cap B = B \cap A$
- **De Morgan's Laws:** $(A \cap B)' = A' \cup B'$
 $(A \cup B)' = A' \cap B'$
- **Associativity:** $A \cup (B \cup C) = (A \cup B) \cup C$
 $A \cap (B \cap C) = (A \cap B) \cap C$
- **Distributivity:** $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$



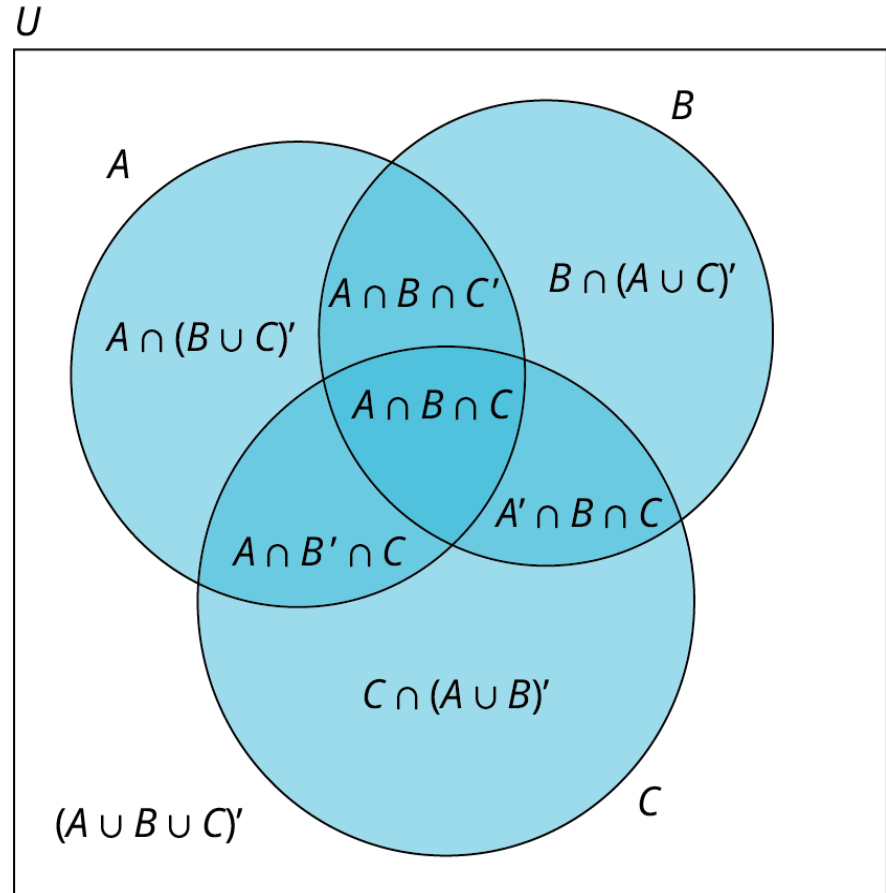
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VENN DIAGRAM WITH TWO INTERSECTING SETS



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VENN DIAGRAMS WITH THREE SETS



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WHAT IS A FUNCTION?

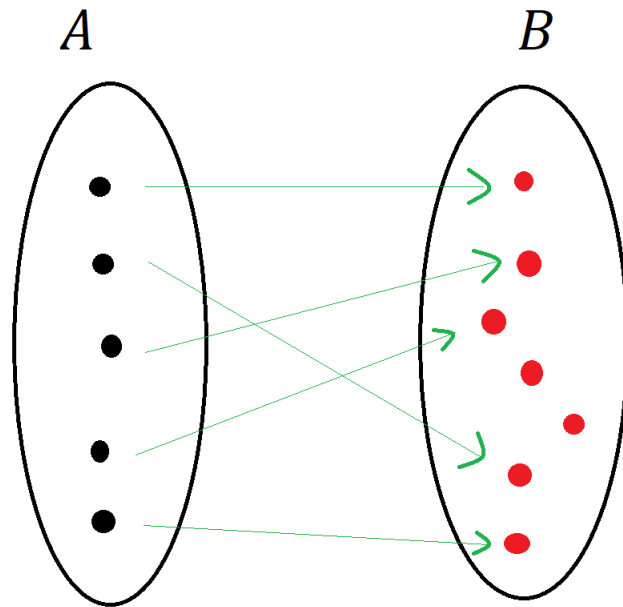
A **function** is a rule for mapping the elements of one set to elements of another set.

We write $f: A \rightarrow B$ to denote a function mapping elements of set A to elements of set B .

A is called the **domain** of f ;

B is called the **range** of f ;

If x is the input of the argument, $f(x)$ is the output or image.



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EXAMPLES

Let $A = \{1, 2, 3, \dots\}$ and $B = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$.

The rule $f: A \rightarrow B$ such that $f(x) = 1/x$ is a function from A to B .

Let $A = \{1, 2, 3, \dots\}$ and $B = \{\dots, -2, -1, 0, 1, 2, \dots\}$.

The rule $f: A \rightarrow B$ such that $f(x) = 2x - 8$ is a function from A to B .



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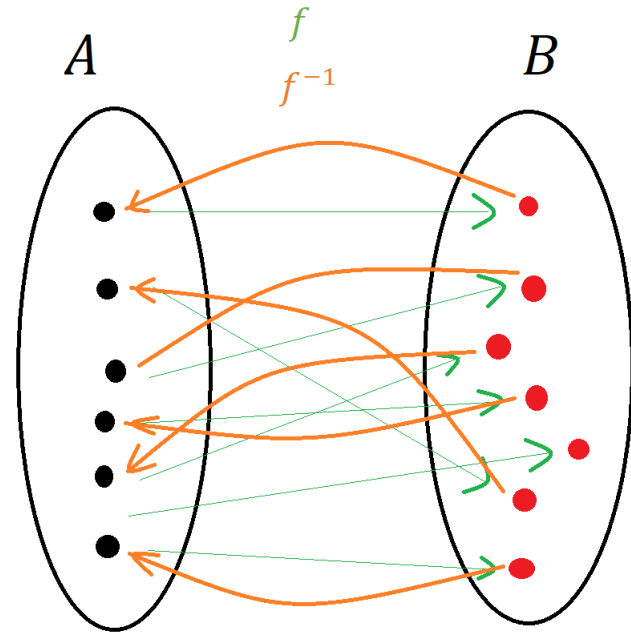
INVERSE OF A FUNCTION

The inverse of a function $f: A \rightarrow B$ is a function $f^{-1}: B \rightarrow A$ such that:

$$f(f^{-1}(y)) = y \text{ for every } y \in B \text{ and}$$
$$f^{-1}(f(x)) = x \text{ for every } x \in A.$$

The inverse of a function does not always exist but when it exists, it is unique.

If $f: \{1, 2, 3, 4\} \rightarrow \{2, 3, 4, 5\}$ is a function such that $f(x) = x + 1$, then its inverse is $f^{-1}: \{2, 3, 4, 5\} \rightarrow \{1, 2, 3, 4\}$ such that $f^{-1}(x) = x - 1$.



QUESTION 1 & SOLUTION

Question: How many subsets do set $A = \{1, 2, 3, 4\}$ have? List all of them.

Solution: The set $A = \{1, 2, 3, 4\}$ has $n = 4$ elements. Therefore, it has $2^n = 2^4 = 16$ subsets.

They are:

$\emptyset, \{1\}, \{2\}, \{3\}, \{4\},$

$\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\},$

$\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\},$

$\{1, 2, 3, 4\}.$

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QUESTION 2

Let $U = \{1, 2, 3, \dots, 10\}$ be the universal set, and subsets A and B are such that $A = \{1, 3, 5, 7, 9\}$ and $B = \{4, 5, 6\}$.

Find $A \cap B$ and $A \cup B$ and $(A \cup B)^c$ and $(A \cap B)^c$.

Find A^c , B^c , $A \cap B^c$, $A^c \cap B$, $A^c \cup B^c$, and $A^c \cap B^c$.

SOLUTION (2)

Given that $A = \{1, 3, 5, 7, 9\}$, $B = \{4, 5, 6\}$, and $U = \{1, 2, 3, \dots, 10\}$, one obtains:

$A \cap B = \{5\}$, $A \cup B = \{1, 3, 4, 5, 6, 7, 9\}$, $(A \cup B)^c = \{2, 8, 10\}$, and $(A \cap B)^c = \{1, 2, 3, 4, 6, 7, 8, 9, 10\}$.

$A^c = \{2, 4, 6, 8, 10\}$, $B^c = \{1, 2, 3, 7, 8, 9, 10\}$, $A \cap B^c = \{1, 3, 7, 9\}$, $A^c \cap B = \{4, 6\}$,

$A^c \cup B^c = \{1, 2, 3, 4, 6, 7, 8, 9, 10\}$, and $A^c \cap B^c = \{2, 8, 10\}$.

QUESTION 3

A panel 12 examiners was to interview two candidates and pick one of them for a job placement. 8 of them said candidate John was suitable, 5 of them said candidate James was suitable, while 2 said neither John nor James was suitable.

- a) Draw a Venn diagram to illustrate this information.
- b) How many said both John and James were suitable?
- c) How many said John alone was suitable?

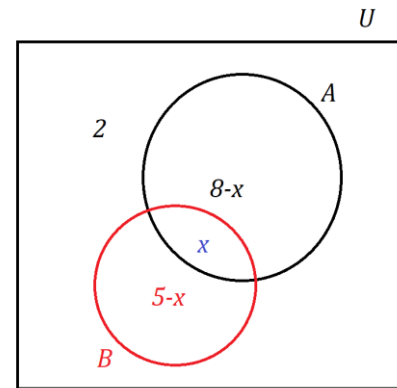
SOLUTION (3)

Let U be the universal set of all 12 examiners.

Let A be the set of examiners that pick John as the suitable candidate.

Let B be the set of examiners that pick James as the suitable candidate.

- a) The Venn diagram herein illustrates this information. Here x is the number of examiners who pick both John and James as suitable candidates: $x = n(A \cap B)$.
- b) Since 2 out of the 12 examiners picked neither John nor James, $n(A \cup B) = 12 - 2 = 10$.
From the formula $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ we have that $n(A \cap B) = n(A) + n(B) - n(A \cup B) = 8 + 5 - 10 = 3$. Thus, $x = 3$ examiners said both John and James were suitable.
- c) The number of candidates who said John alone was suitable is
 $8 - x = 8 - 3 = 5$.



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QUESTION 5

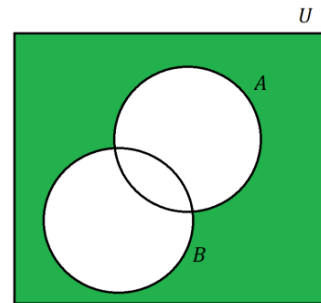
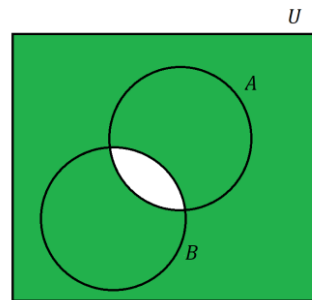
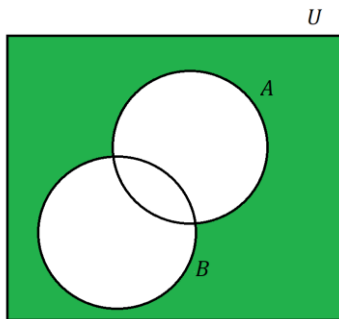
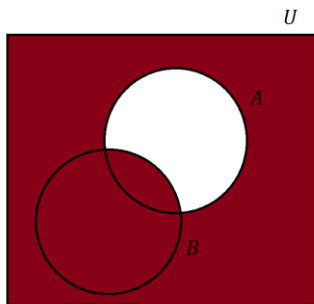
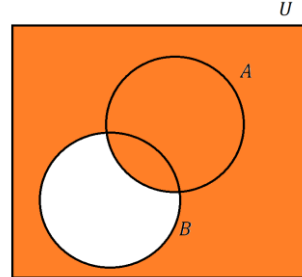
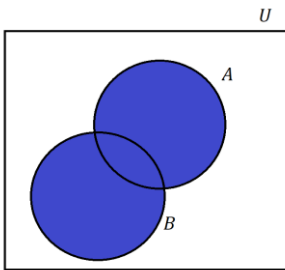
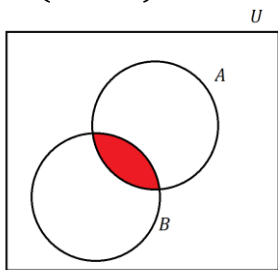
5. Represent the following on a Venn diagram each:
 $A \cap B$; $A \cup B$; $A \cup B^c$; $A^c \cup B$; $A^c \cap B^c$; $A^c \cup B^c$; $(A \cup B)^c$.

QUESTION 4 & SOLUTION

Question: Represent the following on a Venn diagram each:

$A \cap B$; $A \cup B$; $A \cup B^c$; $A^c \cup B$; $A^c \cap B^c$; $A^c \cup B^c$; $(A \cup B)^c$.

Solution:. The following are the representations of the sets $A \cap B$, $A \cup B$, $A \cup B^c$, $A^c \cup B$, $A^c \cap B^c$, $A^c \cup B^c$, and $(A \cup B)^c$ each on a Venn diagram, from left to right.

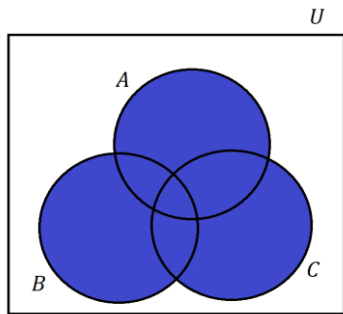


QUESTION 5 & SOLUTION

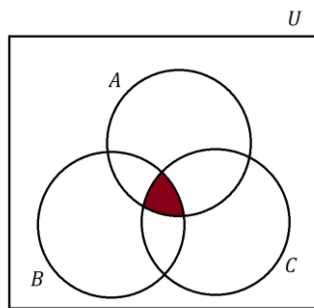
Question: Represent the following on a Venn diagram each:

$A \cup B \cup C$; $A \cap B \cap C$; $(A \cup B \cup C)^c$; $A \cup B \cup C^c$.

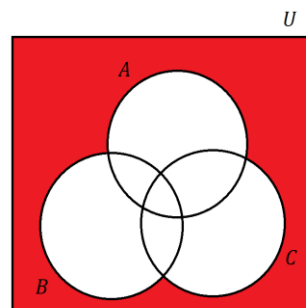
Solution:



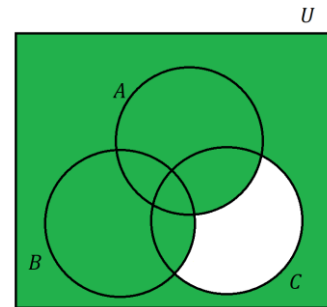
$$A \cup B \cup C$$



$$A \cap B \cap C$$



$$(A \cup B \cup C)^c$$



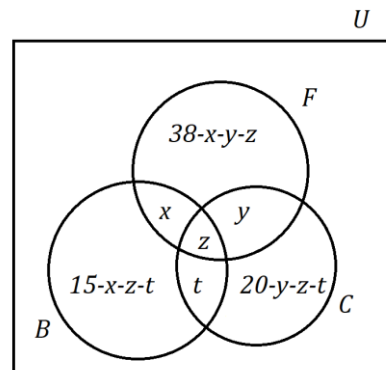
$$A \cup B \cup C^c$$

QUESTION 6

6. The organizers of a men competition awarded 38 medals in Football, 15 in Basketball, and 20 in Cricket. If these medals went to a total of 58 men and only three of these men got medals in all the three sports, how many men received medals in exactly two of the three sports?

SOLUTION (6)

6. Let: U be the universal set of the 58 men awarded medals;
 F the set of the 38 men awarded medals in Football;
 B the set of the 15 men awarded medals in Basketball;
 C the set of the 20 men awarded medals in Cricket.



Men who received medals in exactly two of the three sports belong to either of the sets $F \cap B \cap C^c$, $F \cap B^c \cap C$, or $F^c \cap B \cap C$. Therefore the number of men who received medals in exactly two of the three sports is: $n(F \cap B \cap C^c) + n(F \cap B^c \cap C) + n(F^c \cap B \cap C) = x + y + t$

Only 3 men got medals in all three sports hence $n(F \cap B \cap C) = z = 3$.

All 58 men were awarded medals, hence $n(F \cup B \cup C) = 58$. Therefore:

$$(38 - x - y - z) + (15 - x - z - t) + (20 - y - z - t) + (x + y + t) + z = 58$$

$$67 - (x + y + t) = 58$$

$$x + y + t = 67 - 58 = 9.$$

Therefore, exactly $x + y + t = 9$ men received medals in exactly two of the three sports.

THANK
YOU